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2.1 (a)

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i=5$	$\sum y_i=10$	$\sum(x_i-\bar{x})=0$	$\sum(x_i-\bar{x})^2=10$	$\sum(y_i-\bar{y})=0$	$\sum(x_i-\bar{x})(y_i-\bar{y})=8$

Sample mean: $\bar{x}=1$, $\bar{y}=2$.

(b)
$$b_2=\frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}=\frac{8}{10}=0.8$$

$$b_1=\bar{y}-b_2\bar{x}=2-0.8\cdot 1=1.2$$

(c)
$$\sum_{i=1}^{5}x_i^2=9+4+1+1+0=15$$
$$\sum_{i=1}^{5}x_iy_i=12+4+3-1+0=18$$

$$\left.\begin{array}{l}\sum(x_i-\bar{x})^2=10\\ \sum x_i^2-N\bar{x}^2=15-5\cdot 1^2=10\end{array}\right\}\Rightarrow \sum(x_i-\bar{x})^2=\sum x_i^2-N\bar{x}^2$$

$$\left.\begin{array}{l}\sum(x_i-\bar{x})(y_i-\bar{y})=8\\ \sum x_iy_i-N\cdot\bar{x}\cdot\bar{y}=18-5\cdot 1\cdot 2=8\end{array}\right\}\Rightarrow \sum(x_i-\bar{x})(y_i-\bar{y})=\sum x_iy_i-N\cdot\bar{x}\cdot\bar{y}$$

(d)

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i=5$	$\sum y_i=10$	$\sum\hat{y}_i=10$	$\sum\hat{e}_i=0$	$\sum\hat{e}_i^2=3.6$	$\sum x_i\hat{e}_i=0$

$\hat{y}_i=b_1+b_2x_i=1.2+0.8x_i$, and $\hat{e}_i=y_i-\hat{y}_i$

Sample variance of y: $S_y^2=\sum_{i=1}^{N}\frac{(y_i-\bar{y})^2}{N-1}=\frac{1}{4}(4+1+0+1+4)=2.5$

Sample variance of x: $S_x^2=\sum_{i=1}^{N}\frac{(x_i-\bar{x})^2}{N-1}=\frac{10}{4}=2.5$

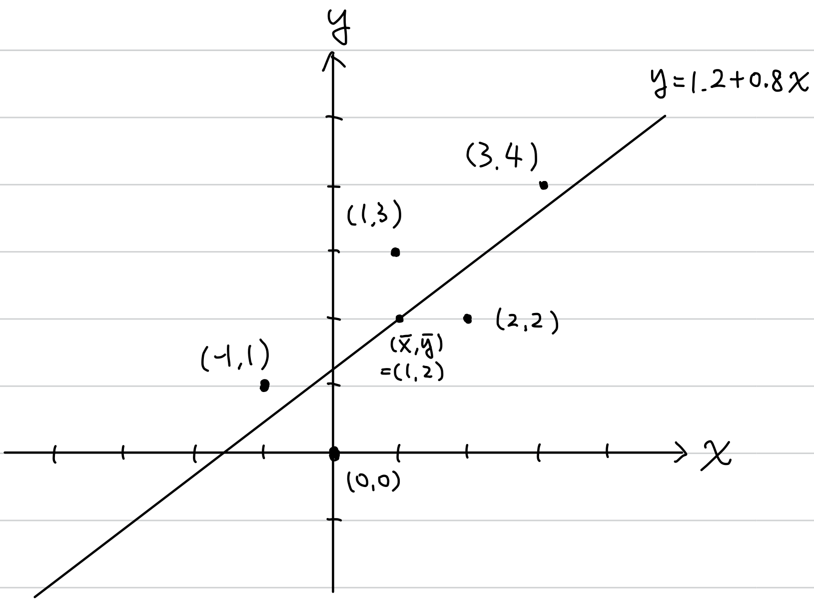
Sample variance between x and y : $S_{xy} = \sum_{i=1}^{N} \frac{(y_i-\bar{y})(x_i-\bar{x})}{N-1} = \frac{8}{4} = 2$

Sample correlation between x and y : $r_{xy} = \frac{S_{xy}}{S_x S_y} = \frac{2}{2.5} = 0.8$

Coefficient of variation of x : $CV_x = 100 \cdot \left(\frac{S_x}{\bar{x}}\right) = 100 \cdot \sqrt{2.5} \approx 158.11\%$

Median of x : 1.

(e)



x	y
3	4
2	2
1	3
-1	1
0	0

(g) $\bar{y} = 2$, $b_1 + b_2\bar{x} = 1.2 + 0.8 \cdot 1 = 2 \Rightarrow \bar{y} = b_1 + b_2\bar{x}$

(h) $\bar{\hat{y}} = \sum \frac{\hat{y}_i}{N} = \frac{1}{5} \cdot 10 = 2 = \bar{y}$.

(i) $\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$

(j) $\widehat{var}(b_2|x) = \frac{\hat{\sigma}^2}{\sum(x_i-\bar{x})^2} = \frac{1.2}{10} = 0.12$

$se(b_2) = \sqrt{\widehat{var}(b_2|x)} = \sqrt{0.12} \approx 0.346$.

2.22

a.

```
> library(POE5Rdata)
> mod1 <- lm(totalscore ~ small, data = star5_small)
> smod1 <- summary(mod1)
> smod1

Call:
lm(formula = totalscore ~ small, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max
-191.63  -50.63   -7.12   42.37  324.38

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  918.628      2.536 362.260   <2e-16 ***
small          9.989      4.611   2.166   0.0305 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 73.36 on 1198 degrees of freedom
Multiple R-squared:  0.003903,	Adjusted R-squared:  0.003071
F-statistic: 4.694 on 1 and 1198 DF,  p-value: 0.03047
```

The estimated coefficient on small (9.989) indicates that student in small classes score about 10 points higher on average than students in regular classes, and this difference is statistically significant at the 5% level ($p=0.0305$).

$$TOTALSCORE_i = 918.628 + 9.989\,SMALL_i + e_i$$

b.

```
> mod2 <- lm(readscore ~ small, data = star5_small)
> smod2 <- summary(mod2)
> smod2

Call:
lm(formula = readscore ~ small, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max
-76.123 -21.123  -3.356  15.877 192.877

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  434.123      1.044 416.021  < 2e-16 ***
small          5.466      1.897   2.881  0.00403 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.19 on 1198 degrees of freedom
Multiple R-squared:  0.006882,	Adjusted R-squared:  0.006053
F-statistic: 8.301 on 1 and 1198 DF,  p-value: 0.004032
```

The estimated coefficient on small (5.466) indicates that student in small classes score about 5.5 points higher in reading on average than students in regular classes, and this difference is statistically significant at the 1% level ($p=0.004$).

$$READSCORE_i = 434.123 + 5.466\,SMALL_i + e_i'$$

```
> mod3 <- lm(mathscore ~ small, data = star5_small)
> smod3 <- summary(mod3)
> smod3

Call:
lm(formula = mathscore ~ small, data = star5_small)

Residuals:
     Min       1Q   Median       3Q      Max
-145.505  -35.028   -6.505   28.495  141.495

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  484.505      1.677 288.884   <2e-16 ***
small          4.522      3.049   1.483    0.138
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48.52 on 1198 degrees of freedom
Multiple R-squared:  0.001832,	Adjusted R-squared:  0.0009992
F-statistic: 2.199 on 1 and 1198 DF,  p-value: 0.1383
```

The estimated coefficient on small (4.522) indicates that student in small classes score about 4.5 points higher in math on average than students in regular classes, but this difference is NOT statistically significant.

$$MATHSCORE_i = 484.502 + 4.522\,SMALL_i + e_i''$$

The estimates suggest that students in small classes perform better in both reading and math. However, the effect is statistically significant for reading scores but not for math scores.

c.

```
> mod4 <- lm(totalscore ~ aide, data = star5_small)
> smod4 <- summary(mod4)
> smod4

Call:
lm(formula = totalscore ~ aide, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max 
-195.14  -52.14   -6.95   40.95  330.86 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  922.145      2.640 349.250   <2e-16 ***
aide          -1.396      4.437  -0.315    0.753    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 73.5 on 1198 degrees of freedom
Multiple R-squared:  8.267e-05,	Adjusted R-squared:  -0.000752 
F-statistic: 0.09904 on 1 and 1198 DF,  p-value: 0.753
```

The estimated coefficient on aide (-1.396) indicates that student in a regular classes with a teacher aide score about 1.4 points lower on average than a class without a teacher aide, but this difference is NOT statistically significant.

$TOTALSCORE = 922.145 - 1.396 AIDE + e$

d.

```
> mod5 <- lm(readscore ~ aide, data = star5_small)
> smod5 <- summary(mod5)
> smod5

Call:
lm(formula = readscore ~ aide, data = star5_small)

Residuals:
    Min      1Q  Median      3Q     Max 
-77.908 -21.536  -2.908  15.464 191.092 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 435.9084     1.0882 400.585   <2e-16 ***
aide         -0.3719     1.8285  -0.203    0.839    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 30.29 on 1198 degrees of freedom
Multiple R-squared:  3.453e-05,	Adjusted R-squared:  -0.0008002 
F-statistic: 0.04137 on 1 and 1198 DF,  p-value: 0.8389
```

The estimated coefficient on aide (-0.3719) indicates that student in a regular classes with a teacher aide score about 0.4 point lower on average in reading than a regular class without a teacher aide, but this difference is NOT statistically significant.

$READSCORE = 435.9084 - 0.3719 AIDE + e'$

```
> mod6 <- lm(mathscore ~ aide, data = star5_small)
> smod6 <- summary(mod6)
> smod6

Call:
lm(formula = mathscore ~ aide, data = star5_small)

Residuals:
     Min       1Q   Median       3Q      Max 
-147.236  -32.236   -7.212   27.788  140.788 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  486.236      1.744 278.731   <2e-16 ***
aide          -1.024      2.931  -0.349    0.727    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 48.56 on 1198 degrees of freedom
Multiple R-squared:  0.0001019,	Adjusted R-squared:  -0.0007327 
F-statistic: 0.1221 on 1 and 1198 DF,  p-value: 0.7268
```

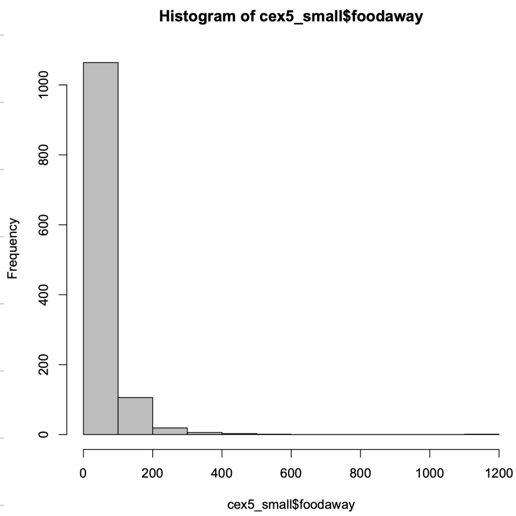
The estimated coefficient on aide (-1.024) indicates that student in a regular class with a teacher aide score about 1 point lower on average in math than a regular class without a teacher aide, but this difference is NOT statistically significant.

$MATHSCORE = 486.236 - 1.024 AIDE + e''$

The estimates suggest that students in a regular class without a teacher aide perform better than a regular class with a teacher aide in both reading and math, but both of them are not statistically significant.

2.25

a.



← Histogram of FOODAWAY

↓ Summary statistics

```
> library(POE5Rdata)
> data("cex5_small")
> hist(cex5_small$foodaway, col='grey')
> summary(cex5_small$foodaway)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   12.04   32.55   49.27   67.50 1179.00
```

mean : 49.27   median : 32.55

25th percentiles : 12.04

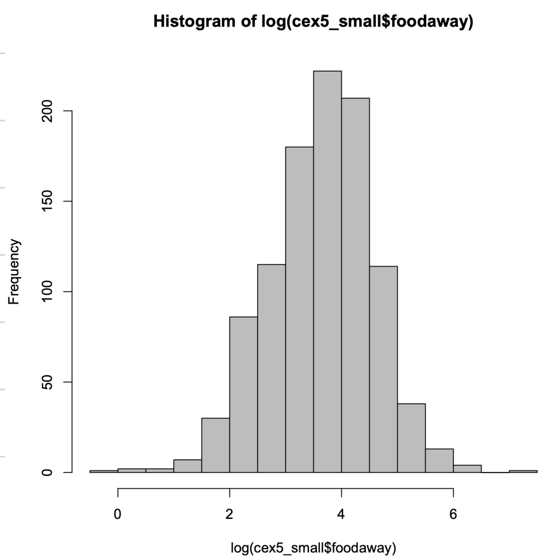
75th percentiles : 67.50

b.

```
summary(cex5_small$foodaway[cex5_small$advanced==1])
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   21.67   48.15   73.15   90.00 1179.00
summary(cex5_small$foodaway[cex5_small$college==1])
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   14.44   36.11   48.60   68.67  416.11
summary(cex5_small$foodaway[cex5_small$college==0 & cex5_small$advanced==0])
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00    9.63   26.02   39.01   52.65  437.78
```

	mean	median
with an advanced degree	73.15	48.15
with a college degree member	48.60	36.11
with no advanced or college degree member	39.01	26.02

c.




```
> summary(log(cex5_small$foodaway))
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   -Inf   2.488   3.483    -Inf   4.212   7.072
```

Since at least one data in FOODAWAY is 0, and the log of 0 doesn't exist, so ln(FOODWAY) has less observation than FOODWAY.

d.

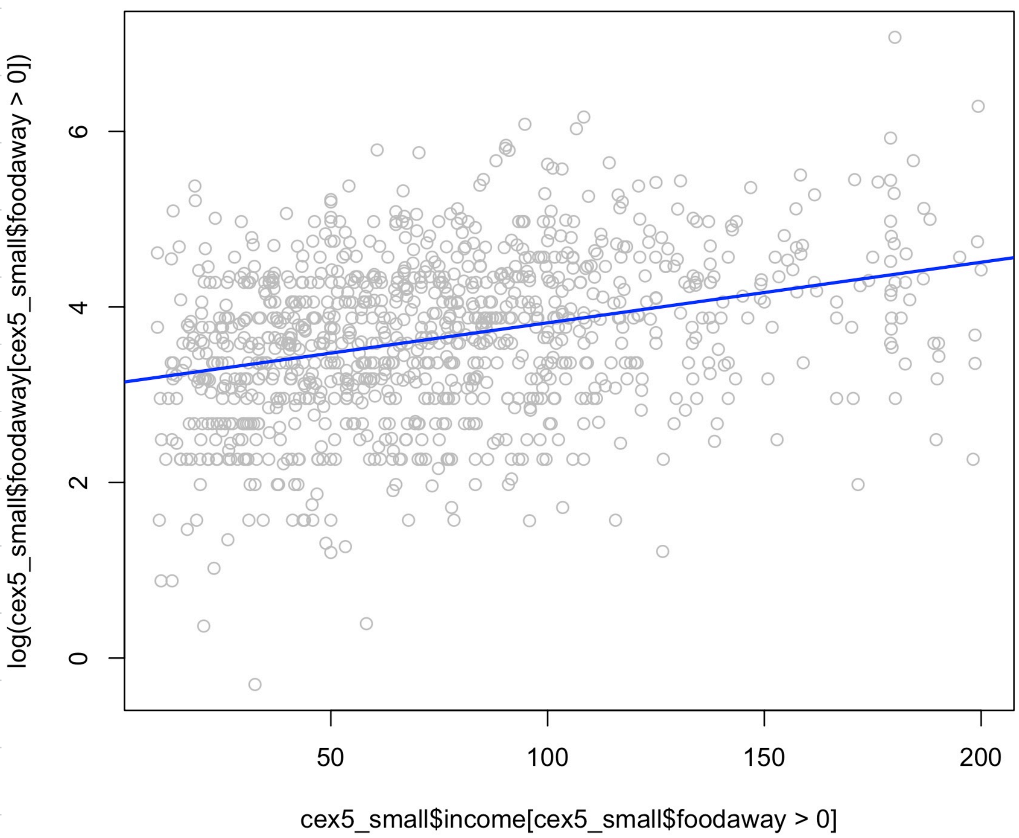
```
> mod1 <- lm(log(foodaway) ~ income, data = cex5_small[cex5_small$foodaway > 0, ])
> smod1 <- summary(mod1)
> smod1

Call:
lm(formula = log(foodaway) ~ income, data = cex5_small[cex5_small$foodaway >
    0, ])

Residuals:
    Min      1Q  Median      3Q     Max
-3.6547 -0.5777  0.0530  0.5937  2.7000

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 3.1293004  0.0565503   55.34   <2e-16 ***
income      0.0069017  0.0006546   10.54   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8761 on 1020 degrees of freedom
Multiple R-squared:  0.09826, Adjusted R-squared:  0.09738
F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

Estimate of the linear regression:

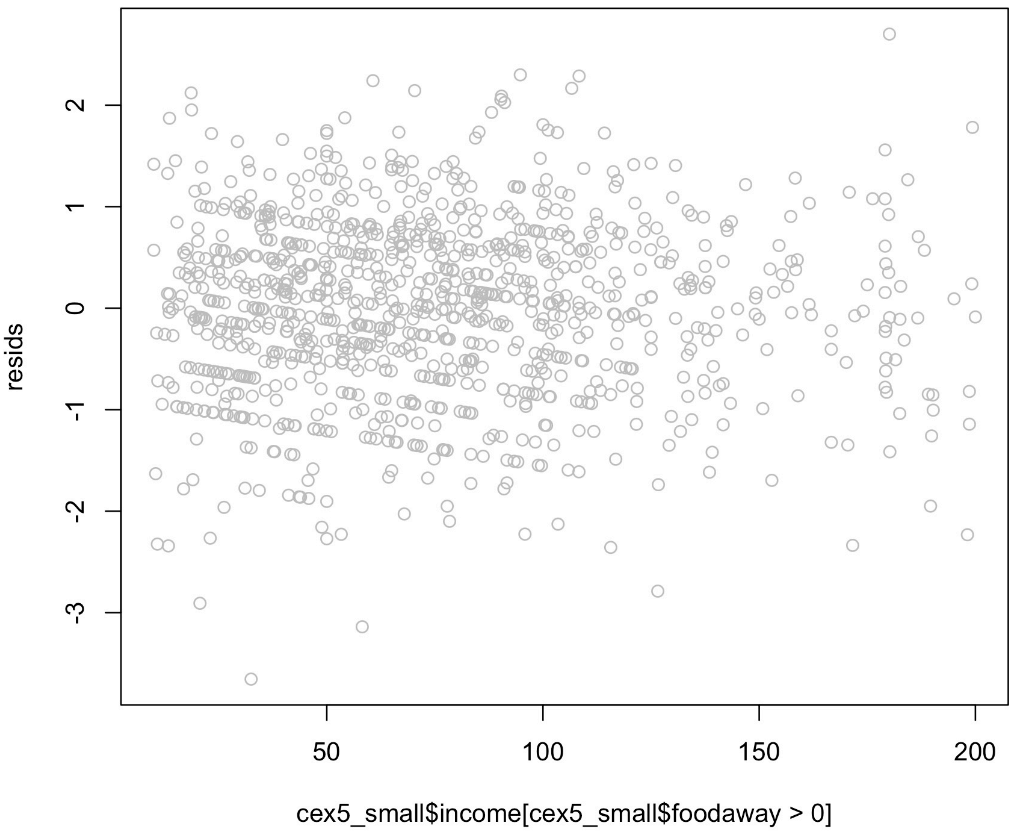
$\ln(FOODAWAY) = 3.1293004 + 0.0069017 \cdot INCOME + e$.

The estimate slope 0.0069017 means that if the monthly income goes up by \$100, the log of the expected food away from home expenditure per month per person will increase approximately by 0.0069017.

e.



f.



The residuals appear to be randomly scattered around zero with no obvious pattern. There's a slight increase in dispersion at higher income levels, suggesting possible mild heteroskedasticity.